A Temporal Receptivity Gate for Xenopus Tail Regeneration: Identifiability Limits and a Falsifiable Mechanistic Hypothesis

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Abstract

Why Xenopus laevis tadpoles temporarily lose tail-regeneration competence remains unresolved at the level of individual animals. We examine whether developmental changes in the timing of basal-epidermal receptivity could uncouple injury signaling from regenerative state induction. The strongest supported result is conditional and analytical: unlinked early and late measurements cannot generally identify individual temporal coupling, even when their stage-specific distributions are known exactly. A restricted model then distinguishes insufficient early ERK activity from delayed epidermal receptivity through opposite predictions for equal-exposure early and late pulses.

These predictions concern modeled transition drive, not demonstrated regeneration rescue. A fixed post-amputation endpoint also confounds pulse timing with response latency, leaving a crucial experimental design obligation. No empirical effect estimate, established novelty, or independent biological confirmation is available. The resulting framework specifies what measurements and counterfactual controls are needed to distinguish a timing gate from signal deficiency without mistaking developmental-stage classification for animal-level prediction.

About this paper

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Evidence coverage limitation

The evidence index omitted 3390 material saved research field fragment(s) across 10 source(s). These fragments were not reviewed for this paper. Interpretations that could depend on omitted material remain provisional; the complete frozen source snapshot and omission ledger are retained with the generation record. *Claims discussed:* C10

1 Introduction and biological context

The target is an explanation of variable regeneration competence in Xenopus laevis, not another association between regeneration and familiar pathways. Basal-epidermal regeneration-initiating cells, abbreviated RICs, and extracellular-matrix remodeling, abbreviated ECM remodeling, are treated as known background under the supplied project description. The proposed contribution is narrower: a stage-dependent temporal dependency between injury signaling and entry into a regenerative epidermal state. Whether this dependency is previously unreported remains unresolved.

A population trajectory and an individual predictive rule answer different questions. Comparing early and late samples across developmental stages may describe average responses, but an animal-level coupling claim requires information connecting an animal's early response to its later state. An explanation of temporary loss and subsequent recovery additionally requires a verified later-competent Xenopus laevis condition; a stage-40 versus stage-46 comparison alone cannot establish the full developmental trajectory.

Claims discussed: C10

2 Methods and evidence

The analysis distinguished supplied biological background from inspected primary evidence. It identified the named GEO accessions as candidate resources but did not establish their sample structure, expression units, replicate counts, pooling, pairing, batches, or regeneration labels. Literature pointers were retained as an incomplete prior-art boundary rather than treated as a completed literature review.

The analytical component defined a stage-conditioned early-to-late coupling coefficient and constructed joint distributions with identical observable marginals but different coupling coefficients. It also examined the regression obtained by assigning stage-level early-response means to late samples, showing why that assignment cannot create an identifiable animal-level interaction.

The mechanistic component formulated restricted signal-amplitude and receptivity-delay models and derived their equal-exposure pulse contrasts. Examination of the proposed experimental interpretation identified response-latency confounding, unvalidated assay capabilities, baseline feasibility, and missing independent biological validation. No transcriptomic reanalysis, predictive benchmark, or intervention experiment was completed in the available evidence.

3 Biological scope and the evidence needed for an individual-level mechanism

An individual-level explanation of tail-regeneration competence in Xenopus laevis must specify a conditional relationship between injury signaling and subsequent regenerative-state induction, rather than merely identify regeneration-associated cells or pathways. The candidate considered here is that developmental stage changes when basal epidermal cells can respond to an injury signal, potentially separating signal availability from the receptive interval for entering a regenerative state. This temporal dependency is a mechanistic hypothesis, not an observed property of the available animals, and its originality remains unassessed.

Its intended contribution is therefore narrower than identifying regeneration-initiating cells (RICs), extracellular-matrix (ECM) remodeling, or familiar injury-signaling pathways as relevant to regeneration. [C2]

The prior-art boundary is substantive: the supplied biological account attributes transient RIC formation from basal epidermal cells and ECM-remodeling defects associated with regeneration failure to the 2024 study identified by DOI 10.1186/s13059-024-03396-3. Those findings are treated as prior knowledge, not discoveries of the present analysis; reactive oxygen species and immune or developmental signaling are likewise excluded from novelty claims. DOI 10.1016/j.ydbio.2021.01.005 and the review DOI 10.3390/ijms252111597 provide additional literature starting points, but their specific experiments and overlap with the proposed dependency have not been established in the supplied evidence.

Consequently, neither these references nor the attributed RIC/ECM findings establish that a stage-dependent response to pulse timing at matched signal exposure is previously unreported; that judgment requires examination of the original and subsequent literature. [C2]

The observational starting point is the project-supplied comparison between stages 40 and 46, associated with GEO GSE245320 and component series GSE245317, GSE245318, GSE245312, and GSE245313. The reported bulk sampling schedule includes 0, 0.5, and 6 hours and 3 days after amputation, but this description does not establish a complete stage-by-time design or coverage by every component assay. The available evidence contains accession pointers rather than an inspected sample manifest, leaving actual biological units, replication, pooling, early-to-late linkage, batches, expression units, file availability, and regeneration-outcome labels unresolved.

These resources belong to the same study and cannot be counted as independent biological confirmations simply because they have separate accession numbers. [C1]

For the individual-level question, the decisive sample information is the relationship between a biological specimen, its molecular measurements, and its eventual regeneration outcome. A usable manifest must distinguish animals from pools and technical libraries, identify stages at both amputation and collection, and resolve collection times, tissue boundaries, treatments, clutch, and processing batch. Technical libraries from one specimen do not add biological replication, while independently collected pools may support pool-level inference without representing measurements from individual animals. These distinctions determine what biological question the observations can address, not merely how many observations enter an analysis. [C1]

The single-cell evidence is described as predominantly characterizing regenerating animals rather than providing matched single-cell observations across competent and refractory conditions. Its appropriate initial role is therefore to inform candidate cell-state definitions, not to supply an assumed refractory counterpart or an assumed link between early and later measurements. Whether any of the named resources supports linked biological-unit analysis remains unresolved: neither individual pairing nor its absence has been established. The evidence needed for an individual-level mechanism must accordingly be distinguished from evidence that can describe developmental-stage differences in populations. [C1]

The biological scope also stops short of explaining the complete loss-and-recovery trajectory. A stage-40/stage-46 contrast, even if its sampling structure is verified, cannot establish why competence subsequently returns; that extension requires an independently verified and sampled later-competent X. laevis condition. Nutritional findings or developmental-stage patterns in X. tropicalis cannot substitute for that condition or establish the proposed mechanism in X. laevis. The resulting evidentiary boundary changes the central question from whether a known regenerative program differs between two stages to whether a specifically timed dependency can explain variation among animals and, ultimately, reverse when competence returns. [C10]

Claims discussed: C1, C2, C10

4 Why unlinked snapshots cannot establish temporal coupling

An animal-level temporal-receptivity hypothesis requires information about how early and late responses vary together, not merely whether their population averages differ between developmental stages. For the proposed Xenopus laevis comparison, let E denote an animal's early wound-response measurement, L its later basal-epidermal regeneration-initiating-cell and extracellular-matrix activation measurement, and S its developmental stage. A descriptive measure of temporal coupling is the within-stage linear-projection slope $\beta_s = \mathrm{Cov}(E, L \mid S = s)/\mathrm{Var}(E \mid S = s)$, provided the required moments exist and $\mathrm{Var}(E \mid S = s) > 0$.

This estimand asks whether animals with stronger early responses subsequently exhibit stronger epidermal activation within the same stage; it does not by itself establish signal transmission or causality. [C3]

The relevant sampling distinction is conditional, because the supplied evidence does not establish whether the named GEO resources contain individual early-to-late links. Biological units, pooling, replication, batches, expression units, sample coverage, pairing, and outcome labels remain unverified for GSE245320 and its component series. The argument below therefore applies if early and late observations come from different, unlinked animals or pools and no additional justified information identifies their joint distribution; it is not a demonstrated description of these accessions. [C1, C3]

Under that sampling premise, separate destructive collections estimate the stage-conditioned marginal distributions $P(E \mid S)$ and $P(L \mid S)$, whereas the coupling slope depends on the joint distribution $P(E, L \mid S)$. Knowing how frequently each early response and each late response occurs does not reveal which early response belongs with which later response. Matching

developmental stage, batch, or clutch does not turn measurements from different animals into repeated measurements of one animal. Even exact knowledge of both marginals can consequently leave the covariance, and hence the coupling slope, undetermined. [C3]

A binary construction makes this limitation explicit. Within one stage, suppose E takes the values -1 and $+1$ with equal probability, so that its mean is 0 and its variance is 1. If the later response is paired as $L = E$, then $\mathrm{Cov}(E, L) = 1$ and the slope is $+1$; if it is paired as $L = -E$, the covariance and slope are instead -1. In either construction, the marginal distribution of L is the same equally probable distribution on $\{-1, +1\}$, so unlinked early and late observations cannot distinguish the two cases. These values are an analytical counterexample, not measured or simulated tadpole expression. [C3]

The same ambiguity extends directly to a developmental-stage interaction. Consider two stage classes, corresponding illustratively to stages 40 and 46, with the preceding early-response distribution in both. One construction assigns $L = E$ at both stages, yielding $\beta_{40} = \beta_{46} = 1$ and a slope contrast $\beta_{46} - \beta_{40} = 0$; another retains $L = E$ at stage 40 but assigns $L = -E$ at stage 46, yielding a contrast of -2. Every stage-specific early and late marginal remains identical across these constructions, although one contains no coupling interaction and the other contains a negative interaction.

Thus neither the existence nor the sign of the proposed individual-level interaction is generally identified by those marginals alone. [C3]

This result is not restricted to populations with equal early and late means. Stage-specific shifts and positive rescaling preserve the possibility of different pairings, so a reproducible stage-by-time expression contrast can coexist with unresolved within-animal coupling. Increasing the number of unlinked samples improves the precision of marginal estimates but cannot distinguish joint distributions that generate identical observables. Likewise, correlating stage/time averages describes a relationship among population summaries rather than estimating β_s. [C3]

Replacing the missing individual early measurement with a stage-specific early mean introduces a separate algebraic failure. Code the two stages as $S \in \{0, 1\}$ and assign each late sample the predictor $\widetilde{E} = e_0 + (e_1 - e_0)S$, where e_0 and e_1 are the respective early means. Because $S\widetilde{E} = e_1 S$, the intercept, stage, assigned early predictor, and stage-by-predictor interaction span at most two dimensions: the proposed regression cannot separately identify its response and interaction coefficients. Additional late biological replicates can sharpen estimates of stage means, but they cannot repair this rank deficiency or convert the assigned mean into an individual predictor. [C4]

The identifiability boundary therefore separates a potentially estimable population trajectory from the animal-level dependency required by the temporal-receptivity argument. It does not preclude descriptive differential expression, nor does it refute biological coupling; actual individual links, informative interventions, or justified structural restrictions could change what is identifiable. Linked pools would still support a pool-level analysis rather than automatically establishing animal-level prediction. The consequence for the central argument is substantive: a developmental contrast in early and late activation cannot serve as evidence of individual temporal uncoupling unless the missing joint information is supplied. [C3]

Claims discussed: C1, C3, C4

5 A restricted temporal-receptivity model and its discriminating prediction

A restricted temporal-receptivity model makes the proposed developmental gate distinguishable from insufficient injury signaling by predicting opposite responses to early and late ERK supplementation. For Xenopus laevis basal epidermis, the hypothesis is that stage-46 cells remain inducible but become receptive later than stage-40 cells, reducing the overlap between injury-associated ERK activity and the capacity to enter a transient regenerative epidermal state. This is a mechanistic candidate, not an observed developmental shift; its discriminating feature is the timing of signal effectiveness at matched added activity, rather than the presence of ERK activity or RIC-state induction alone [C5].

Let $E_s(t)$ denote basal-epidermal ERK activity to be measured at developmental stage s and post-amputation time t, and let $C_s(t)$ denote hypothetical receptivity to enter the transient regenerative epidermal state. Define cumulative transition drive as $J_s = k \int E_s(t)C_s(t)\,dt$, with $k > 0$. In this model, activity contributes to induction only to the extent that it overlaps a receptive epidermal state: the same activity delivered at different times can therefore contribute differently to transition drive. Receptivity is a latent model quantity, not an established molecular measurement, and its relationship to an operationally defined epidermal state remains to be determined [C5].

The signal-deficiency alternative, A_{amp}, places receptivity in an early window at both stages but attributes stage-46 failure to insufficient early ERK activity. The delayed-receptivity alternative, B_{delay}, retains early receptivity at stage 40 while shifting stage-46 receptivity into a later window. Both alternatives can produce inadequate induction of a transient RIC-like state, so a broad description such as "failure to induce RICs" does not distinguish them. Their contrast instead concerns whether the missing contribution is early signal amplitude or temporal overlap between signal and receptivity [C5].

To expose that distinction, consider nonnegative added-activity pulses $p_{\text{early}}(t)$ and $p_{\text{late}}(t)$ confined to separate early and late windows, with equal integrated exposure: $\int p_{\text{early}}(t)\,dt = \int p_{\text{late}}(t)\,dt = Q > 0$. Candidate experimental windows are 0.5–2 hours and 4.5–6 hours post-amputation; these are prospective choices, not measured receptive periods. The analytical comparison holds the untreated within-stage activity trajectory and receptivity fixed while adding either pulse, and assumes no independent timing-dependent injury or growth effect [C5, C6].

Under those restrictions, the untreated trajectory cancels from the stage-46 late-minus-early contrast, giving $\Delta J_{46} = J_{46,\text{late}} - J_{46,\text{early}} = k \int C_{46}(t)[p_{\text{late}}(t) - p_{\text{early}}(t)]\,dt$. This cancellation is what makes the comparison informative: the contrast depends on where the added activity falls relative to receptivity, rather than on a difference in total added activity. It does not require the endogenous activity trajectory itself to be identical between developmental stages, because the subtraction is performed within stage 46 [C6].

In the idealized signal-deficiency model, stage-46 receptivity equals a constant $c > 0$ throughout the early pulse window and is zero throughout the late pulse window. Consequently, $\Delta J_{46} = -kcQ < 0$: early supplementation contributes transition drive, whereas late supplementation does not. Reversing those receptive supports in the delayed-receptivity model gives $\Delta J_{46} = +kcQ > 0$,

so late supplementation contributes more than early supplementation. These opposite signs are analytical consequences of the stipulated receptive windows, not biological effect estimates [C6].

The sign result is narrower than a general distinction between signal deficiency and timing failure. For nonrectangular or overlapping receptive windows, the integral measures the actual pulse-weighted difference in receptivity; merely labeling a mechanism "delayed" does not guarantee a positive contrast. Equal exposure removes a total-added-activity difference, but it does not replace the requirement that the pulses overlap regions with different receptivity. Thus, the exact discriminating prediction belongs to the two restricted temporal-support models, not to every possible mechanism within the broader categories of injury-signaling failure or RIC/ECM failure [C6].

Finally, transition drive is not equivalent to tail regrowth. Transferring the predicted sign to a regrowth contrast requires a monotone, nonsaturated downstream response and no separate dominant block; saturation or an independent downstream defect can erase the distinction in regrowth, while absent receptivity in both tested windows erases it at the level of transition drive itself. The model therefore changes the mechanistic question from whether more ERK activity promotes regeneration to whether equal added activity has opposite timing preferences under two specified epidermal-state hypotheses. That conditional timing contrast, rather than supplementation alone, is the proposed discriminator [C5, C6].

Diagram (readable relationship list). The proposed temporal-receptivity model separates injury-associated ERK activity from a hypothetical epidermal receptive window. Their overlap determines modeled transition drive. Connections to RIC-state induction, ECM remodeling, and regrowth are hypotheses or downstream assumptions, not demonstrated causal results.

Node 1. Developmental stage in Xenopus laevis

Node 2. Basal-epidermal ERK activity

Node 3. Hypothetical temporal receptivity

Node 4. Modeled signal–receptivity overlap

Node 5. RIC-state induction

Node 6. ECM remodeling

Node 7. Functional tail regeneration

Edge 1. Node 1 points to node 2: Restricted amplitude alternative

Edge 2. Node 1 points to node 3: Restricted timing alternative

Edge 3. Node 2 points to node 4: Signal input

Edge 4. Node 3 points to node 4: Temporal weighting

Edge 5. Node 4 points to node 5: Proposed induction relationship

Edge 6. Node 5 points to node 6: Proposed downstream relationship

Edge 7. Node 6 points to node 7: Unvalidated downstream translation

Boundary 1. Restricted mechanistic hypothesis; not an observed causal pathway: node 1, node 2, node 3, node 4, node 5, node 6, node 7

Evidence: [Research session 21, artifact 4]; claims: C5, C6, C8, C9

Claims discussed: C5, C6

6 From pulse contrast to a valid counterfactual test

A valid counterfactual test of temporal receptivity must distinguish an effect of wound age at stimulation from an effect of the time available to respond after stimulation. In the proposed stage-40 and stage-46 Xenopus laevis comparison, collecting all molecular readouts at six hours post-amputation does not make those response intervals comparable: an early pulse necessarily precedes collection by longer than a late pulse. Consequently, a stage-dependent early-versus-late contrast at that endpoint cannot, by itself, identify a wound-age-dependent receptivity gate. This is a limitation of the proposed experimental contrast, not an observed property of tadpole response kinetics [C7].

Transient induction and subsequent decay provide a concrete alternative explanation for an apparent late-pulse advantage. At the common endpoint, an early treatment could already have induced a response that has declined, whereas a late treatment could be measured nearer its response maximum. If induction or decay kinetics also differ between developmental stages, this alternative can produce a stage-by-pulse-timing interaction without the proposed shift in receptivity. The relevant comparison must therefore address response trajectories, rather than treating a larger terminal molecular score as evidence of greater induction [C7].

Matched-latency, vehicle-adjusted molecular measurements are necessary before interpreting the timing interaction mechanistically. Early- and late-pulse responses must be compared at corresponding elapsed times after treatment, with vehicle measurements accounting for the untreated trajectory at the relevant stage and wound age. These comparisons must be incorporated into a schedule capable of distinguishing differential inducibility from differences in response onset, persistence, and decay; merely adding serial collections does not establish that distinction.

A completed schedule meeting these requirements has not been specified, so the six-hour endpoint remains insufficient rather than becoming a selective test through an assumed correction [C7].

Exposure matching imposes a separate requirement: the manipulated quantity must be verified basal-epidermal ERK activity, not nominal illumination duration or administered dose. A light-gated RAF/ERK actuator is a proposed implementation, whose expression specificity, activation kinetics, reversibility, and noninjurious illumination would first require validation in X. laevis basal epidermis. Early and late perturbations would need comparable measured amplitudes and activity integrals under a prespecified tolerance.

If endogenous early activity is near saturation, increasing the early light dose does not establish comparable added activity; failure to achieve matched perturbations would preclude interpreting the efficacy contrast as a timing effect [C8].

The complementary blockade experiment likewise requires demonstrated target engagement and reversibility. A reversible MEK inhibitor is an implementation candidate, but local delivery alone would not establish basal-epidermal specificity: vehicle, washout, and epidermal versus

non-epidermal response measurements are required. If only systemic inhibition is feasible, a stage-40 blockade result would concern ERK necessity somewhere in the animal, not necessity specifically within basal epidermis. Moreover, stage-46 rescue combined with stage-40 blockade would not independently identify a timing gate; the exposure-matched early-versus-late stimulation contrast remains essential to the proposed discrimination [C8].

RIC-state induction must also be operationally distinguishable from signal delivery and from later regrowth. The proposed measurements include induction per basal-epidermal cell, absolute epidermal abundance, and subsequent ECM remodeling, but the state assay and its sensitivity remain to be established. Rescue without a detected state change would challenge the specified mediation model only if the assay sampled the relevant cells and times with adequate sensitivity. Otherwise, an undetected or short-lived transition could be mistaken for its absence, and an increase in total molecular signal could be mistaken for a within-cell state change [C8].

Controls must rule out treatment differences that would change the meaning of the timing contrast. Wounded vehicle controls, light-only actuator-negative controls, actuator-positive no-light controls, and unamputated treatment controls are proposed to distinguish pathway actuation from illumination, toxicity, and developmental effects. Measurements of wound size, closure, epidermal viability, and tissue composition are needed to determine whether apparent state loss instead reflects altered injury or cell depletion.

A favored pulse must also improve over vehicle: a late-versus-early advantage caused only by early-treatment damage is not rescue, while regrowth through generic growth effects would not establish the proposed epidermal mediation [C8].

Spatial localization remains an alternative explanation even when activity totals and cell abundances are comparable. The supplied analytical construction shows that identical marginal injury and ECM activities can coexist with different local alignment; it does not demonstrate that such arrangements occur in tadpoles or determine regeneration. Spatial measurements could therefore inform the interpretation, but they are not yet a validated control or independent confirmation: reporter specificity, temporal registration, anatomical gradients, and tissue-mixture effects remain unresolved.

In particular, reassigning measurements within anatomical strata can alter spatial autocorrelation and patch structure, so an apparent alignment effect under that null would not automatically isolate a biological coupling mechanism [C11].

These requirements change what the pulse experiment can establish. A difference at six hours would show a treatment-timing-dependent endpoint response, but attribution to wound-age-dependent basal-epidermal receptivity requires comparable achieved exposure, latency-resolved vehicle contrasts, and a sufficiently sensitive state assay. The mechanistic test therefore turns on whether differential state induction survives those counterfactual comparisons, rather than on the presence of a favorable terminal score alone [C7, C8].

Claims discussed: C7, C8, C11

7 Validation thresholds and the unresolved loss-and-recovery trajectory

Validation of the temporal-receptivity hypothesis requires separating a developmental population contrast, an animal-level predictive association, and a perturbation-supported mechanism. These conclusions answer different questions: whether molecular responses differ between conditions, whether measurements predict subsequent outcomes in unseen animals, and whether selectively changing signal timing alters the proposed regenerative transition. The supplied evidence contains no empirical effect estimate, uncertainty interval, held-out baseline comparison, or independent biological confirmation establishing the latter two conclusions; even the archival population contrasts depend on unresolved sample structure.

The evidentiary thresholds below therefore define what would justify each conclusion rather than reporting validation already achieved [C1, C9].

Confirmatory analysis must first fix the biological unit of inference and establish which developmental contrasts the sampling design can distinguish. Independent animals and independently collected pools must be identified explicitly, with pool sizes, technical libraries, clutch membership, and early-to-late linkage retained in the analysis structure; a pool cannot silently become an individual-animal observation. Stage, collection time, batch, expression units, and outcome availability must also be established before selecting a model.

If stage and batch cannot be distinguished, the corresponding developmental effect is not identifiable; if early and late measurements are unlinked, analysis must remain at the level of supported population contrasts rather than animal-level temporal prediction. Different component accessions from the same study do not, by themselves, provide independent validation [C1].

The molecular definitions and prediction target must likewise be fixed before confirmatory outcomes are examined. This includes injury-response and RIC/ECM module membership, normalization, explicit handling of Xenopus laevis homeolog mappings, and procedures for separating tissue-composition changes from changes within the relevant epidermal population. Prediction of later molecular activation and prediction of functional regeneration should be treated as distinct tasks, each with a defined measurement time, outcome, and scoring procedure.

In particular, a molecular endpoint cannot serve as its own predictor, and an assay requiring destructive collection cannot be treated as a longitudinal measurement in an animal subsequently followed for regrowth [C1, C9, C11].

A predictive claim would require improvement over stage-only, RIC/ECM-only, and known-signaling baselines under a preregistered primary metric and biological-unit evaluation splits. The relevant question is whether the proposed timing rule adds information about subsequent outcomes beyond those simpler descriptions, not merely whether it separates developmental stages. Evaluation must therefore distinguish between-stage discrimination from predictive information among animals within a stage, where sufficient outcome variation exists.

A RIC/ECM-only baseline for individual regeneration prediction would require a validated live surrogate or an explicitly narrower feasible comparison; measurements from separate destructive cohorts cannot substitute for the missing paired baseline [C9, C11].

Uncertainty procedures must be specified alongside the estimands rather than appended after a favorable comparison. A separate pilot should inform the minimum biologically meaningful effect, between-animal and between-clutch variation, measurement error, outcome imbalance, and the confirmatory sample size; the proposed enrollment alone does not establish adequate precision. Module selection, baseline models, outcome scoring, multiplicity handling, and interval-based decision rules should then be fixed before confirmatory collection.

For both predictive improvement and perturbation effects, failure to reach statistical significance is inconclusive when the interval still includes a meaningful benefit: exclusion requires adequate precision relative to a prespecified effect-size bound, together with valid exposure and assay checks [C9].

Spatial organization remains a consequential alternative explanation rather than an independent validation of the temporal gate. The supplied analytical construction shows that identical marginal injury and ECM activity can coexist with different spatial alignment, so abundance adjustment alone does not determine whether the relevant activities meet locally. However, reassigning measurements within anatomical strata can change spatial autocorrelation and patch structure, allowing shared anatomical gradients to resemble specific alignment.

Any spatial sensitivity analysis would therefore need validated reporters, reliable registration, explicit exchangeability assumptions, and checks on retained spatial structure; the current proposal does not yet supply a calibrated negative control or biological confirmation [C11].

Mechanistic interpretation requires more than successful prediction or a pulse-associated increase in regrowth. Independent randomized molecular and regeneration measurements must connect the selective perturbation to epidermal state induction while addressing injury severity, wound closure, viability, generic growth, developmental progression, metabolic state, and tissue composition. A downstream-bypass comparison is needed before attributing an observed coupling defect specifically to signal transmission rather than an inability to enter the regenerative state.

Conversely, rescue without a detected state transition would challenge the proposed mediation only if the assay adequately covered the relevant cells and times and had sufficient sensitivity to detect that transition [C9, C11].

Originality is a separate threshold from either predictive or causal support. RIC identity, basal-epidermal origins, ECM-remodeling defects, and familiar injury-signaling pathways are excluded from the proposed contribution. The potentially distinctive claim is the conditional timing relationship, but its novelty remains unassessed because the cited original publications and subsequent literature have not been inspected for that specific interaction. A species-specific account of the experiments and dependencies already tested is therefore necessary before describing the timing rule as previously unreported, regardless of whether future experiments support it [C2].

Finally, a stage-40 versus stage-46 comparison cannot establish why competence later returns. Extending the hypothesis to the complete developmental trajectory requires an independently verified later-competent X. laevis condition, followed by measurements and perturbation tests that determine whether the same timing rule accounts for competence there. Nutritional findings or stage patterns in Xenopus tropicalis cannot supply that missing condition or establish the X. laevis mechanism. The explanatory target is thus not a classifier of two stages, but a rule that survives

independent animal-level prediction and selective perturbation tests across both the loss and the recovery of competence [C9, C10].

Table 1: Evidence thresholds for distinct future conclusions about the temporal-receptivity hypothesis. These are requirements for interpretation, not achieved validation results.

Conclusion	**Minimum supporting evidence**	**Inference not justified by that evidence alone**
Descriptive population contrast	Verified biological units, pooling, stage/time coverage, expression units, replication, and an identifiable stage–batch design.	An individual early-to-late coupling rule or prediction of an animal's regeneration outcome.
Independently supported predictive association	Linked measurements and outcomes; fixed prediction times and scoring; independent biological evaluation against stage-only, feasible RIC/ECM-only, and known-signaling baselines; prespecified metrics and uncertainty.	A causal timing gate, signal-transmission defect, or explanation of later regained competence.
Perturbation-supported mechanism	Independent randomized molecular and regeneration evidence from validated selective interventions, with exposure and assay checks, alternative-explanation controls, and precision-based effect criteria.	Established novelty or an explanation of the full loss-and-recovery trajectory without prior-art assessment and a verified later-competent X. laevis test.

Evidence: [Research session 21, artifact 1], [Research session 21, artifact 3], [Research session 21, artifact 4]; claims: C1, C2, C9, C10, C11 *Claims discussed:* C1, C2, C9, C10, C11

8 Findings and hypotheses

8.1 C1 – Limitation

The actual biological units, sample coverage, replication, pairing, batches, expression units, and outcome labels of the named GEO resources remain unverified in the supplied evidence. The available records explicitly distinguish accession pointers and the reported sampling schedule from an

inspected sample manifest. Thus no actual replicate count, matched-animal structure, or phenotype availability can be asserted. The accessions are described as components of one study, not as independent biological replications. *Evidence:* [Research session 21, artifact 5], [Research session 21, artifact 3]

8.2 C2 – Unresolved

The temporal-receptivity hypothesis has unassessed novelty; RIC identity, basal-epidermal origins, ECM defects, and familiar injury-signaling pathways are not claimed as discoveries. The project attributes RIC and ECM findings to DOI 10.1186/s13059-024-03396-3. DOI 10.1016/j.ydbio.2021.01.005 and DOI 10.3390/ijms252111597 are additional literature pointers, but their detailed contents and subsequent literature were not inspected. These records establish a project-level exclusion from novelty, not a completed prior-art search. *Evidence:* [Research session 21, artifact 1], [Research session 21, artifact 4]

8.3 C3 – Established

If only unlinked stage-conditioned early and late marginal distributions are observed, without additional identifying restrictions, individual temporal coupling and its developmental-stage interaction are not generally identifiable. Let E denote early response, L later activation, and S stage, with $\beta_s = \mathrm{Cov}(E, L \mid S = s)/\mathrm{Var}(E \mid S = s)$ when the denominator is positive. In the saved binary construction, E takes -1 and $+1$ equally often. Pairing $L = E$ gives slope $+1$, whereas pairing $L = -E$ gives slope -1, while preserving both marginals. Using the first pairing at both stages gives a slope contrast of 0; switching the second stage to the opposite pairing gives -2 without changing any stage-specific marginal. The construction establishes the conditional analytical limitation, not its applicability to the actual accessions or independent formal verification. *Evidence:* [Research session 21, artifact 1], [Research session 21, artifact 2], [Research session 21, artifact 5]

8.4 C4 – Established

Assigning each late sample its stage's early-response mean does not recover an individual predictor and produces a rank-deficient stage-by-response regression in the two-stage example. For stage coding $S \in \{0, 1\}$, the assigned predictor is $E = e_0 + (e_1 - e_0)S$, and its interaction is $SE = e_1 S$. Consequently, the intercept, stage, assigned predictor, and interaction span at most two dimensions. Additional late samples can improve estimation of stage means but cannot repair this algebraic redundancy. *Evidence:* [Research session 21, artifact 2]

8.5 C5 – Conjecture

A stage-dependent shift in basal-epidermal receptivity to injury-associated ERK activity is a specific mechanistic candidate, distinct from insufficient early signal amplitude. The proposed model defines basal-epidermal ERK activity $E_s(t)$, receptivity $C_s(t)$, and cumulative transition drive $J_s = k \int E_s(t) C_s(t)\, dt$, with $k > 0$. The amplitude alternative retains early receptivity at both

stages but posits insufficient early activity at stage 46. The delay alternative retains early receptivity at stage 40 and shifts stage-46 receptivity to a later window. These are deliberately restricted alternatives; broad failure to induce RICs includes both. *Evidence:* [Research session 21, artifact 4]

8.6 C6 – Established

Within the idealized restricted model, equal-exposure late and early pulses produce opposite signs of the stage-46 transition-drive contrast under signal deficiency and delayed receptivity. For nonnegative pulses p_{early} and p_{late} with equal integral $Q > 0$, untreated activity cancels in $\Delta J_{46} = k \int C_{46}(t)[p_{\text{late}}(t) - p_{\text{early}}(t)]\, dt$. If receptivity equals $c > 0$ only in the early window, the contrast is $-kcQ$; reversing the receptive support gives $+kcQ$. These are analytical values, not measured effects. Nonrectangular receptive windows require the actual pulse-weighted receptivity contrast, and translation to regrowth requires a monotone, nonsaturated downstream response without a dominant independent block. *Evidence:* [Research session 21, artifact 4]

8.7 C7 – Limitation

A common six-hour post-amputation molecular endpoint confounds pulse timing with elapsed response time and therefore cannot alone identify a wound-age-dependent receptivity gate. Early and late pulses have different latencies before a fixed endpoint. Transient induction, decay, or stage-dependent response kinetics can therefore mimic a pulse-timing interaction. Matched-latency, vehicle-adjusted measurements are required, but a complete schedule separating wound age from post-pulse time is not supplied. *Evidence:* [Research session 21, artifact 5], [Research session 21]

8.8 C8 – Unresolved

Selective basal-epidermal perturbation, exposure matching, and a valid RIC-state readout remain feasibility and operational-definition obligations. The saved proposals require a validated actuator, reversible blockade, tissue-specific response measurements, and a defined regenerative-state assay. Rescue without measured state induction challenges mediation only if the assay adequately covers the relevant cells and times. Unequal exposures, toxicity, wound closure, or composition changes would prevent a selective timing interpretation. *Evidence:* [Research session 21, artifact 4], [Research session 21, artifact 5], [Research session 21, artifact 1]

8.9 C9 – Limitation

Neither a new biological mechanism nor a reproducible predictive association is established because empirical effects, uncertainty estimates, baseline comparisons, and independent biological confirmation are absent. The evidence records no empirical effect or independent validation. Confirmatory inference would require biological-unit splits, stage-only, RIC/ECM-only, and known-signaling baselines, a primary metric, multiplicity handling, and uncertainty accounting for biological and clutch variation. A nonsignificant comparison without adequate precision cannot falsify the proposed

protocol. *Evidence:* [Research session 21], [Research session 21, artifact 1], [Research session 21, artifact 4]

8.10 C10 – Limitation

The candidate is restricted to Xenopus laevis and does not yet explain later regained competence. The records require independent verification and sampling of a later-competent Xenopus laevis condition. Xenopus tropicalis nutritional findings or stage patterns are not evidence for the proposed Xenopus laevis mechanism. *Evidence:* [Research session 21, artifact 1], [Research session 21, artifact 2], [Research session 21, artifact 4]

8.11 C11 – Unresolved

Spatial localization and cell composition are relevant alternative explanations, but the supplied spatial proposal does not yet provide a validated control or independent confirmation. The saved spatial construction shows that identical marginal activity totals can coexist with different injury–ECM alignment, so abundance alone does not determine alignment. However, its proposed reassignment null can alter spatial autocorrelation and patch structure. Reporter specificity, registration, anatomical gradients, and baseline measurement feasibility remain unresolved; this proposal cannot serve as validation of the temporal gate. *Evidence:* [Research session 21, artifact 3]

9 Limitations

- Actual accession coverage and biological linkage remain unknown, so the conditional nonidentifiability result cannot yet be attributed to the GEO samples themselves.
- No expression matrices, animal-level outcomes, fitted effects, uncertainty intervals, predictive comparisons, or intervention outcomes are supplied.
- The temporal model compares deliberately restricted alternatives. A positive pulse response would not by itself distinguish every possible failure of RIC induction, metabolic gating, or developmental progression.
- The sign contrast is derived for cumulative transition drive. Its interpretation as a molecular-state or regeneration contrast requires additional downstream assumptions and valid temporal measurements.
- The full wound-age versus post-pulse-latency sampling design, selective perturbation capability, and RIC-state assay remain unresolved.
- Novelty has not been established by inspected primary literature and later-publication coverage.
- A verified later-competent Xenopus laevis condition and independent biological confirmation are absent; no cross-species transfer is justified.

10 Open obligations

- Acquire a row-per-library manifest for GSE245320, GSE245317, GSE245318, GSE245312, and GSE245313, recording species, stage, time, assay, units, animals or pools, biological and technical replication, batches, pairing, accessibility, and phenotype labels.
- Determine which contrasts are identifiable. If early and late measurements are unlinked, stop animal-level temporal-coupling estimation from those snapshots and retain only justified population contrasts.
- Inspect the cited original studies and subsequent literature through the investigation date, including the conflicting late-October and early-November 2026 coverage targets recorded in the evidence, and establish a precise literature cutoff before any novelty claim.
- Validate tissue-specific activity measurement, pulse delivery, reversible blockade, exposure matching, state-score normalization, and assay coverage in Xenopus laevis.
- Complete a molecular sampling schedule that separates wound age from elapsed post-pulse time, including matched-latency and vehicle-adjusted measurements.
- Operationalize RIC-state induction and distinguish it from ECM activity, general wound closure, tissue growth, viability, and changing cell composition.
- Specify reciprocal perturbation and downstream-bypass tests, with sham, vehicle, wound-severity, toxicity, and developmental-progression controls.
- Use a separate pilot to determine a minimum-interest effect, biological and clutch variance, measurement error, and a justified confirmatory sample size; freeze these choices before testing.
- Preregister biological-unit splits, baseline models, predictor availability, primary endpoints, multiplicity handling, and uncertainty procedures. Do not substitute separate destructive cohorts for linked individual predictors.
- Obtain independent unseen-specimen molecular and functional regeneration evidence, and independently verify a later-competent Xenopus laevis condition before claiming an explanation of recovery.

11 Frozen obligation reconciliation

Unlisted obligations are not presumed resolved. The archived research snapshot retains complete source references and the full reconciliation.

- ChatGPT plan or app usage limit reached. Manage usage at https://chatgpt.com/settings/usage; no API-key fallback occurred. (HTTP 400) No strategy was saved. *(Not listed by author; not presumed resolved; 1 saved source reference(s))*

- No research artifact was saved for this strategy. *(Not listed by author; not presumed resolved; 3 saved source reference(s))*
- Acquire original GEO sample metadata and files; verify species, exact stages/times, independent animals versus pools, pairing, batches, replicate counts, outcome labels and assay availability before analysis. If only unlinked snapshots exist, stop the animal-level temporal-coupling analysis. *(Not listed by author; not presumed resolved; 3 saved source reference(s))*
- Inspect the cited original papers and later literature through the investigation date; record exact evidence for or against prior reporting of the pulse-timing interaction. If already established, abandon its novelty claim. *(Not listed by author; not presumed resolved; 3 saved source reference(s))*
- Validate basal-epidermal exposure and response measurements, noninjurious pulse delivery and stable response-score normalization. If early and late exposures cannot be matched, stop rather than interpret their contrast as a timing effect. *(Not listed by author; not presumed resolved; 3 saved source reference(s))*
- Use a separate pilot to justify the minimum-interest margin, estimate between-animal and between-clutch variance, and choose a finite confirmatory sample size for the interaction. Do not select the response module or effect threshold using confirmatory outcomes. *(Not listed by author; not presumed resolved; 3 saved source reference(s))*
- Obtain independent randomized molecular and regeneration results with uncertainty. Resolve cell composition, developmental progression, metabolic state and injury-severity alternatives; a pulse rescue alone does not identify the decoder. *(Not listed by author; not presumed resolved; 3 saved source reference(s))*
- Animal-level prediction and any claim explaining later regained competence require additional longitudinal measurements and a verified later-competent X. laevis condition, respectively. *(Not listed by author; not presumed resolved; 3 saved source reference(s))*
- Audit original GEO metadata, biological units, pairing, pools, batches, phenotypes and file availability. *(Not listed by author; not presumed resolved; 4 saved source reference(s))*
- Inspect original and subsequent literature for exact overlap with the pulse-timing interaction. *(Not listed by author; not presumed resolved; 4 saved source reference(s))*
- Validate pulse delivery, local exposure, basal-epidermal assays and normalization; separate wound age from post-pulse response latency. *(Not listed by author; not presumed resolved; 4 saved source reference(s))*
- Justify the minimum-interest margin and determine confirmatory sample size using separate pilot animals and clutch-level variability. *(Not listed by author; not presumed resolved; 4 saved source reference(s))*
- Obtain independent randomized molecular and regeneration results, with uncertainty and composition, metabolic, developmental and injury controls. *(Not listed by author; not presumed resolved; 4 saved source reference(s))*

- Require individual measurement–outcome links for animal-level prediction and a verified later-competent condition for regained-competence claims. *(Not listed by author; not presumed resolved; 4 saved source reference(s))*
- Material and recoverable: the fixed 6-hour endpoint confounds pulse timing with elapsed time since treatment. Stage-dependent response kinetics can produce the proposed interaction without the claimed wound-age-dependent gate. *(Not listed by author; not presumed resolved; 2 saved source reference(s))*
- Material and recoverable: neither accession structure nor prior reporting of the timing interaction has been inspected. The mathematical limitation cannot yet be attributed to the actual GEO samples, and novelty cannot be claimed. *(Not listed by author; not presumed resolved; 2 saved source reference(s))*
- Material and recoverable: no independent molecular or regeneration measurements support the candidate, and no verified later-competent X. laevis condition tests regained competence. These are missing evidence, not refutations. *(Not listed by author; not presumed resolved; 2 saved source reference(s))*
- The proposed timing contrast confounds wound age with response latency. *(Not listed by author; not presumed resolved; 2 saved source reference(s))*

113 further distinct obligations are retained in the archived reconciliation.

12 Conclusions

The supported analytical conclusion is conditional: unlinked early and late marginals do not generally identify an individual temporal-coupling rule, and assigning stage means cannot supply the missing linkage.

A basal-epidermal temporal-receptivity gate provides a specific hypothesis with opposite modeled pulse-contrast signs under restricted alternatives, but the experimental interpretation remains unresolved because timing, latency, and selective manipulation must be separated.

The evidence does not establish a novel predictor or biological mechanism for the temporary loss and return of tail regeneration. That claim requires completed prior-art assessment, numerical effects with appropriate baselines and uncertainty, and independent biological confirmation across the relevant Xenopus laevis conditions.

Claims discussed: C3, C4, C5, C6, C7, C8, C9, C10

A Selected source-linked research excerpts

The following are bounded, selected excerpts of original research records. They are not exhaustive; consult the full frozen snapshot archived with this paper. They do not constitute independent verification.

A.1 Research excerpt 1

Source and field: [Research session 21, artifact 4]: skeptic_output/reasoning_assessments

[{"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "The boundary matches the project-supplied description and explicitly distinguishes that description from inspected literature. It does not claim verified novelty or independent replication.", "step_number": 1}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "The equations define coherent restricted candidate models.

They are not inferred from data, and the argument correctly recognizes that broad RIC-induction failure contains both alternatives.", "step_number": 2}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "With C and the untreated trajectory fixed, linearity of the integral gives the stated contrast. Constant c on only one pulse window yields exactly -kcQ or +kcQ.

Transfer to regrowth is explicitly conditional on the listed monotone, nonsaturated response assumption; it does not follow from the R and M differential equations alone.", "step_number": 3}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "This is a feasibility-gated intervention proposal, not an assertion that a validated X. laevis reagent exists.

Measuring activity rather than equating illumination durations appropriately addresses stage-dependent actuator response.", "step_number": 4}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "Under early-supported stage-40 receptivity, suppressing relevant early activity reduces cumulative transition drive whereas late suppression...

A.2 Research excerpt 2

Source and field: [Research session 21]: outcome/solution_review/strategy_findings/1/reasoning_steps

[{"argument": "Let E_i denote an animal's early wound-response measurement, R_i its later epidermal RIC/ECM-transition measurement, and S_i developmental stage. An animal-level coupling estimand is the within-stage slope Cov(E,R|S)/Var(E|S), when the variance is positive. Measurements from different animals at different times estimate marginal distributions, not this joint distribution. Whether the actual GEO samples provide animal linkage, independent animals, pools, or only technical/library replication remains unverified.

Pool-level results must not be represented as animal-level prediction.", "basis_artifact_ids": [], "epistemic_status": "deduction", "number": 1, "research_artifact_id": "[Research session 21, artifact 2]", "review": {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "Cov(E,R|S)/Var(E|S) is a defined within-stage linear-projection slope when the relevant moments exist and variance is positive. Unlinked sampling supplies marginals rather than animal correspondence. The step explicitly leaves the GEO sampling premise unverified."}, "title": "Define the estimand and the unresolved sampling prerequisite"}, {"argument": "Within one stage, let U be uniform on [0,1].

Model I assigns E=U and R=U; Model II assigns E=U and R=1-U. Both models give exactly the same uniform marginal distributions for E and R. However, Var(E)=1/12, while Cov(E,R)=1/12 in

Model I and -1/12 in Model II; the coupling slopes are therefore +1 and -1. Independently sampled early and late animals have identical observable distributions under these two...

A.3 Research excerpt 3

Source and field: [Research session 21, artifact 1]: skeptic_output/reasoning_assessments

[{"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "The step explicitly treats the supplied biological background as a project constraint and distinguishes reference pointers from inspected sources. This accepts a stated premise and provenance boundary, not the underlying literature claims as independently established facts.", "step_number": 1}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "The proposed regression slope depends on a joint covariance, whereas separate marginal samples do not generally determine that covariance. Matching group labels does not create individual pairing.

The slope is meaningful where its requisite moments exist and early-response variance is positive.", "step_number": 2}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "For symmetric ś1 early responses, variance is 1. Setting the later response equal to or opposite the early response gives covariance and slope +1 or -1 while preserving both marginals. The two stated stage contrasts are therefore exactly 0 and -2. This is an analytic construction, not computation or frog data.", "step_number": 3}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "Stage-specific shifts and positive rescaling leave alternative pairings possible.

Group-mean correlations consequently do not recover within-stage individual covariance. Cell-state annotation alone likewise supplies no missing longitudinal identity or refractory sampling.", "step_number": 4},...

A.4 Research excerpt 4

Source and field: [Research session 21, artifact 2]: investigator_output/reasoning_steps

[{"argument": "Let E_i denote an animal's early wound-response measurement, R_i its later epidermal RIC/ECM-transition measurement, and S_i developmental stage. An animal-level coupling estimand is the within-stage slope Cov(E,R|S)/Var(E|S), when the variance is positive. Measurements from different animals at different times estimate marginal distributions, not this joint distribution. Whether the actual GEO samples provide animal linkage, independent animals, pools, or only technical/library replication remains unverified.

Pool-level results must not be represented as animal-level prediction.", "basis_artifact_ids": [], "epistemic_status": "deduction", "title": "Define the estimand and the unresolved sampling prerequisite"}, {"argument": "Within one stage, let U be uniform on [0,1]. Model I assigns E=U and R=U; Model II assigns E=U and R=1-U. Both models give exactly the same uniform marginal distributions for E and R. However, Var(E)=1/12, while Cov(E,R)=1/12 in Model I and -1/12 in Model II; the coupling slopes are therefore +1 and -1. Independently sampled early and late animals have identical observable distributions under these two models. Thus no analysis of those unlinked measurements alone can distinguish even the sign of individual coupling.

Additional unlinked time points do not supply the missing individual correspondence without extra, testable structural assumptions. This construction is an identifiability proof, not a biological model of tadpole expression.", "basis_artifact_ids": [], "epistemic_status": "deduction", "title": "Prove nonidentifiability from...

A.5 Research excerpt 5

Source and field: [Research session 21, artifact 3]: skeptic_output/reasoning_assessments

[{"basis_artifact_ids": [], "disposition": "unresolved", "objection": "This entry is designated an assumption, while claim_is_conditional is false. It cannot be accepted as an established premise. Its accession descriptions and prior-art attributions remain background awaiting inspection.", "rationale": "The evidence boundary is appropriately disclosed, and the project-supplied references are correctly treated as provisional rather than inspected sources.", "step_number": 1}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "Direct arithmetic gives mean(I)=mean(M)=1/2, mean(I*M_A)=1/2, and mean(I*M_B)=0, hence covariances 1/4 and -1/4.

The conclusion is correctly restricted to non-identification by the specified marginals.", "step_number": 2}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "Acceptance concerns the definition only. Biological inference still requires a defensible exchangeability model and sensitivity to spatial autocorrelation, registration error, and stratum choice.", "rationale": "T minus its specified null expectation is a coherent definition of excess alignment.

The text explicitly limits interpretation to the chosen constraints and does not claim reporter validity or control of all anatomy.", "step_number": 3}, {"basis_artifact_ids": [], "disposition": "unresolved", "objection": "Neither the positive association nor its developmental loss-and-return pattern follows from the illustrative construction. They require biological observations; adjustment alone cannot establish a coupling defect.",...

A.6 Research excerpt 6

Source and field: [Research session 21, artifact 5]: skeptic_output/reasoning_assessments

[{"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "The supplied context contains accession references but no original sample tables, matrices or publications. Treating sample structure and pairing as unverified follows from the available evidence.", "step_number": 1}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "The centered variables take values ś1 with equal probability. Same-sign pairing gives covariance +1, opposite-sign pairing gives -1, and early-response variance is 1 in both cases.

Swapping the assignments between stages reverses the slope interaction while preserving each marginal.", "step_number": 2}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "Observational equivalence proves nonidentifiability in the unrestricted marginal-only model. Increasing unlinked sample counts does not recover the missing joint distribution.

The stated qualifications concerning longitudinal links, structural restrictions and pool-level versus

animal-level inference are appropriate.", "step_number": 3}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "The ledger accurately distinguishes project-supplied descriptions from inspected literature and makes no priority claim. It is a provisional source list, not a completed novelty audit.", "step_number": 4}, {"basis_artifact_ids": [], "disposition": "challenged", "objection": "At 6 hours, early and late pulses have different response latencies. Transient induction or score decay can generate...

A.7 Research excerpt 7

Source and field: [Research session 8]: steps/0/provider_input_audit/request_body/text/format/schema/propert

{"items": {"$ref": "#/$defs/ResearchReasoningStep"}, "maxItems": 24, "title": "Reasoning Steps", "type": "array"}

A.8 Research excerpt 8

Source and field: [Research session 21]: steps/9/output/reasoning_assessments

[{"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "The boundary matches the project-supplied description and explicitly distinguishes that description from inspected literature. It does not claim verified novelty or independent replication.", "step_number": 1}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "The equations define coherent restricted candidate models.

They are not inferred from data, and the argument correctly recognizes that broad RIC-induction failure contains both alternatives.", "step_number": 2}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "With C and the untreated trajectory fixed, linearity of the integral gives the stated contrast. Constant c on only one pulse window yields exactly -kcQ or +kcQ.

Transfer to regrowth is explicitly conditional on the listed monotone, nonsaturated response assumption; it does not follow from the R and M differential equations alone.", "step_number": 3}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "This is a feasibility-gated intervention proposal, not an assertion that a validated X. laevis reagent exists.

Measuring activity rather than equating illumination durations appropriately addresses stage-dependent actuator response.", "step_number": 4}, {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "Under early-supported stage-40 receptivity, suppressing relevant early activity reduces cumulative transition drive whereas late suppression...

A.9 Research excerpt 9

Source and field: [Research session 21, artifact 2]: objective_review/argument_steps

[{"argument": "Let E_i denote an animal's early wound-response measurement, R_i its later epidermal RIC/ECM-transition measurement, and S_i developmental stage. An animal-level coupling

estimand is the within-stage slope Cov(E,R|S)/Var(E|S), when the variance is positive. Measurements from different animals at different times estimate marginal distributions, not this joint distribution. Whether the actual GEO samples provide animal linkage, independent animals, pools, or only technical/library replication remains unverified.

Pool-level results must not be represented as animal-level prediction.", "basis_artifact_ids": [], "epistemic_status": "deduction", "number": 1, "research_artifact_id": "[Research session 21, artifact 2]", "review": {"basis_artifact_ids": [], "disposition": "accepted", "objection": "", "rationale": "Cov(E,R|S)/Var(E|S) is a defined within-stage linear-projection slope when the relevant moments exist and variance is positive. Unlinked sampling supplies marginals rather than animal correspondence. The step explicitly leaves the GEO sampling premise unverified."}, "title": "Define the estimand and the unresolved sampling prerequisite"}, {"argument": "Within one stage, let U be uniform on [0,1].

Model I assigns E=U and R=U; Model II assigns E=U and R=1-U. Both models give exactly the same uniform marginal distributions for E and R. However, Var(E)=1/12, while Cov(E,R)=1/12 in Model I and -1/12 in Model II; the coupling slopes are therefore +1 and -1. Independently sampled early and late animals have identical observable distributions under these two...

6 further research records are retained in the archived source snapshot.

Evidence and authorship note

Snapshot SHA-256: 3236a8c66a96d713f28726271169e67c88c8a768f398ab3b9b504fee3cdcc74e Generated by an AI author from the cited saved research snapshot. This document is not independent verification. No external bibliography was added.

Validation warnings

- Evidence indexing omitted 3390 material saved research field fragment(s). Those fragments were not reviewed for this paper; the interpretation is provisional where the omitted evidence may matter. Consult the frozen source snapshot and omission ledger for full details.
- Several top-level claim-reference arrays use zero-padded identifiers, whereas the claim register uses unpadded identifiers. These links require correction before this plan is used to generate an article.
- Section-purpose text is incomplete in this draft and must be completed before article generation; the intended proof explanations, control comparisons, and validation discussion must not be inferred from truncated sentences.
- The index is a bounded preview with substantial omitted material, not an exhaustive semantic review. No unseen fragment has been treated as inspected evidence.
- The analytical counterexamples are illustrative mathematical constructions, not observed expression values or biological experiments. They have not undergone independent formal

verification.

- The evidence does not establish whether the actual GEO collections satisfy the unlinked-sampling premise.
- The prior-art ledger is provisional. The named DOI pointers do not support detailed literature conclusions without inspection.
- The fixed-endpoint timing objection remains material; the present record does not contain a complete corrected sampling schedule.
- An archived redox-persistence proposal had an additional identifiability objection: its persistence statistic was proportional to the exposure integral, so integral matching fixed the proposed statistic. That proposal is not used as corroboration of the ERK timing model.
- The evidence index explicitly omitted 3390 saved field fragment(s); see the generation coverage ledger and review the complete archived snapshot.
- Structured paper indexing selected 447 source-linked text fragments from 15 archived sources without AI extraction calls. The complete frozen snapshot remains archived; this bounded index does not constitute review of every saved field.
- C3 is labeled established, but cited saved evidence includes unverified, incomplete, or conjectural material.
- C3 is labeled established while the frozen evidence contains counterexample or contradiction language; inspect the cited sources before accepting it.
- C4 is labeled established, but cited saved evidence includes unverified, incomplete, or conjectural material.
- C4 is labeled established while the frozen evidence contains counterexample or contradiction language; inspect the cited sources before accepting it.
- C6 is labeled established, but cited saved evidence includes unverified, incomplete, or conjectural material.
- C6 is labeled established while the frozen evidence contains counterexample or contradiction language; inspect the cited sources before accepting it.
- The frozen evidence contains open proof/research obligation(s) requiring reconciliation: 121 item(s) are not listed by the author and are not presumed resolved; 10 author-listed item(s) have no exact frozen-record match. See the obligation reconciliation.
- The methods narrative has no linked claim IDs; check it for unsupported factual statements.
- A diagram's graph could not be laid out without likely crossings or overlap; its complete nodes, boundaries, and relationships are shown in a readable text representation.

Typesetting verification note

Needs review: TeX verification is incomplete. The authored research content and safe accepted typesetting corrections were retained. Review the saved proofreading diagnostics before relying on the layout.